Section 2.1: Systems of Linear Systems by the Echelon Method

Finite Mathematics MTH 132 Spring 2014

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Johnson C. Smith University

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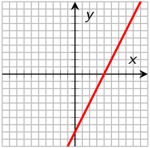

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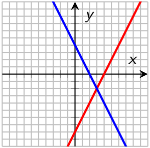

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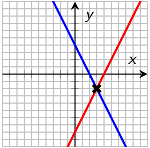

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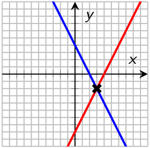

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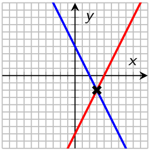

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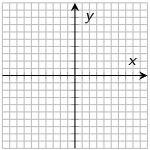

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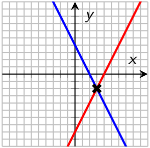

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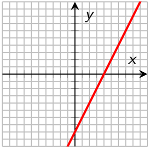

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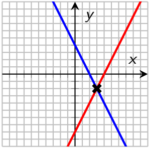

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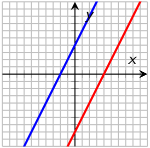

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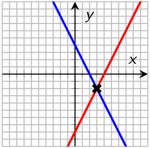

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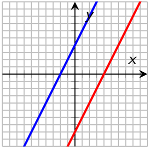

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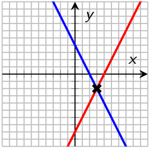

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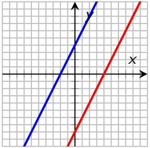

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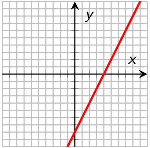

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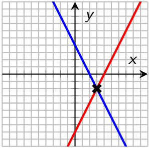

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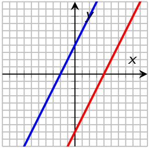

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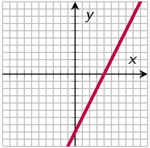

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Use the echelon method to solve the system $y = 2x - 8$ and $y = -2x + 4$

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step 1

Write the system aligning the x's, y's, and constants along columns

step 2

Eliminate one of the variables, and solve.

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step 3

Substitute the solution from *step 2* to find the remaining variable.

Example 2.1.2.

Solve the system

$$\begin{array}{rcrcr} -3x & + & y & = & 4 \\ 2x & - & 2y & = & -4 \end{array}$$

Example 2.1.3.

Solve the system

$$\begin{array}{rcrcl} 9x & - & 5y & = & 1 \\ -18x & + & 10y & = & 1 \end{array}$$

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Example 2.1.4.

A shop sells skirts for \$45 and pants for \$35. Its entire stock is worth \$51,750. During their big sale, half the skirts and two-thirds of the pants were sold for \$30,600. How many skirts and pants are left in stock?

Example 2.1.5.

Solve the system

$$\begin{aligned} 45p &+ 35s &= 51,750 \\ \tfrac{2}{3}45p &+ \tfrac{1}{2}35s &= 30,600 \end{aligned}$$

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Answer: $(0, 2)$